

Figmin XR: AR content creation platform

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ABSTRACT

Figmin XR is a spatial computing application that allows non-technical users to easily create, collect & play with digital content.

Featuring:

- Built-in search & import of 3D models, animated GIFs and videos provided by Sketchfab, Giphy & YouTube;
- Voxel editor for creating 3D and 2D models;
- Text label editor;
- Physics editor;
- 3D model Voxelizer;
- An inventory system to save and load content from;
- Import & export;
- Hand physics for a realistic interaction with physics enabled content.

Figmin XR is a multi-purpose app for the consumer AR market. It runs on Magic Leap 1, Hololens 2 and Nreal light.

See it in action:

<https://youtu.be/0z3P21WLNiU>

CCS CONCEPTS

• ; • **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools; User interface programming; Human computer interaction (HCI); Interaction paradigms; Mixed / augmented reality; Human computer interaction (HCI); Interactive systems and tools; User interface programming; Interaction design; Interaction design process and methods; Contextual design; Interaction design; Interaction design process and methods; User interface design;

KEYWORDS

Art, Design, Content creation, Dynamics/Simulation, Interaction, UI/Tools, VR/AR/MR, Physics modeling, Hand tracking, Hand physics

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1 INTRODUCTION

With Figmin XR anyone can create a mixed reality experience.

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Figure 1: Mini-golf mixed reality scene, created & played on FigminXR



Figure 2: Discover panel

The easiest way to get started is on the Discover Panel. Here you can search and import Animated GIFs, 3D models and videos directly into the app, no modeling skills required.

Objects can be placed anywhere in the room and saved to the user's collection.

Advanced users can create 3D and 2D models with our built-in Voxel editor, featuring user-generated brushes.

Each model can optionally have physics, we have a fully-featured physics editor that even allows the creation of gravitational fields.

Sketchfab 3D models can be voxelized within the app, to simplify their geometry and memory requirements. They can later be fully edited with our Voxel editor.

Once a model has had its physics tuned, it can be interacted with through our hand physics mechanics, or from far away with our ray-based interaction system.



Figure 3: Voxel editor

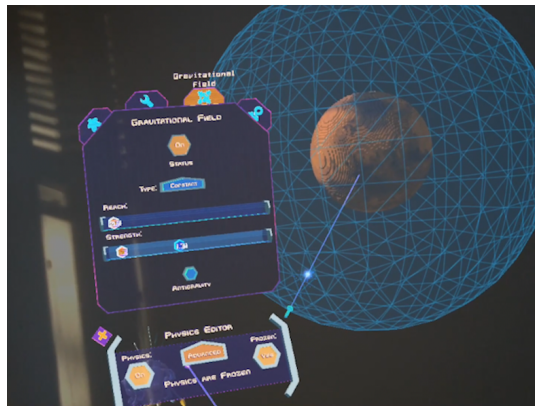


Figure 4: Creating gravitational field with physics editor



Figure 5: Hand physics example

Note that what we mean by hand physics is that we have a physically enabled hand skeleton that applies forces to digital objects it comes in contact with, just like your real hand as it comes into contact with real objects.

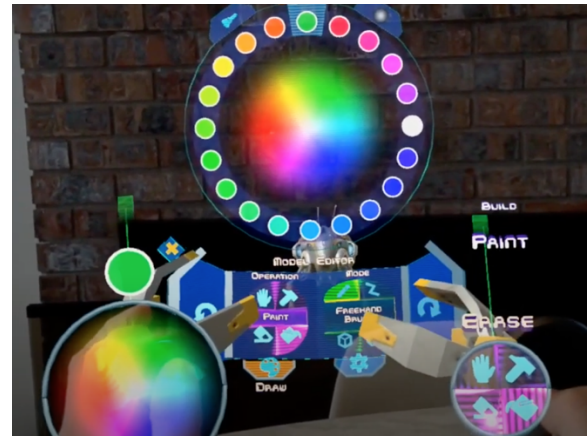


Figure 6: Example of Control Hand User Interface

Entire scenes can be saved to the collection and shared with others via web-links or QR codes.

2 DESIGN & IMPLEMENTATION

The development of Figmin XR began in 2016 with the launch of Hololens 1, as a research project into spatial computing. In 2019 Mr. Davalos was selected as a recipient of a major grant by Magic Leap and our company Overlay was incorporated.

Due to the age of the project, most of the software was designed and programmed from scratch.

In spatial computing, ideally, the interface should be 3D, so a great deal of development time went into this to create all the functionality and components required to build the app: buttons, toggles, tabs, sliders, widgets, etc.

When designing the UX it was a requirement to build it in such a way that the application could be used with motion controllers, hand tracking and on mobile phones.

The main menu is presented by a robot assistant, to avoid the need of binding it to the users' hands. This frees up the hands for our Contextual Hand User Interface (CHUI). It is a control mechanic we designed to facilitate operating on the items the user is working on.

The content shown on the hand by the CHUI depends on the context of what is being done. If an object is being grabbed, a transform widget is shown, if a model is pointed at by the hand ray, an options menu can be recalled to save it, remove it or edit it.

Widgets from the main menu can be equipped to the users' hand, for example while using the voxel editor a user could equip the color palette on one hand, and the undo/redo buttons on the other.

CONCLUSION

Figmin XR is a true spatial computing platform that aggregates different editors and online services into a single cohesive and easy-to-use app. It has novel control mechanics and it is the first AR software to feature hand physics.